SOFTENG 700 · PROJECT 102

How much performance does an AI scheduler lose when 5G traffic changes?

Testing whether the AI needs retraining for each type of traffic

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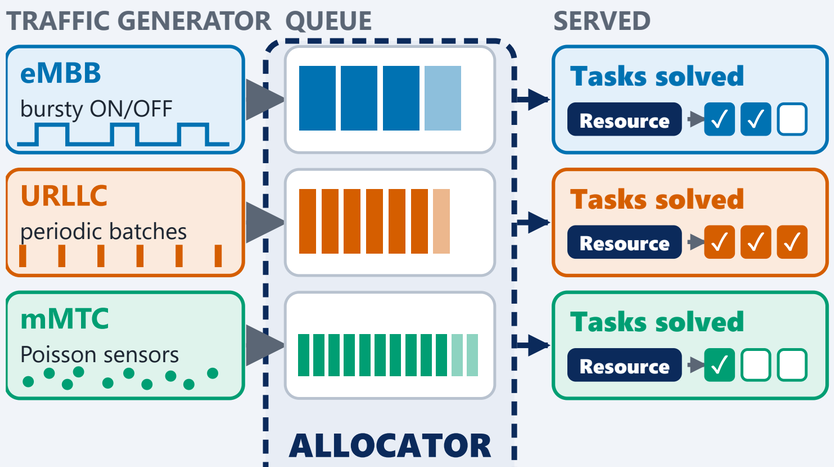
1 The problem

Every millisecond, a 5G base station must share its limited radio capacity between three types of task:

- **eMBB** video and downloads
- **URLLC** robots and cars
- **mMTC** sensor readings

Reinforcement learning (RL) is widely proposed to optimise this decision.

2 Simulator & four environments



Environment	Load share (%)
Balanced	33 / 33 / 33
eMBB-intensive	50 / 25 / 25
URLLC-intensive	25 / 50 / 25
mMTC-intensive	25 / 25 / 50

3 How the AI learns

+ **REWARD** when tasks in each task type are finished on time.

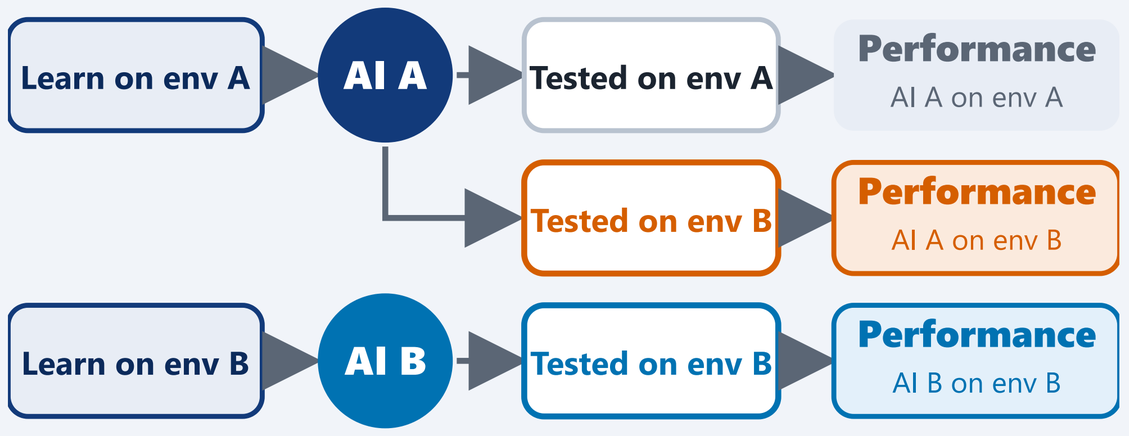
– **PENALTY** when tasks pile up and wait in the queue.

Learns by trial and error

Practice 5 separate runs per traffic mix

Length 600 rounds in each run

4 What has been measured



Performance loss
AI B performance minus AI A performance, both on env B

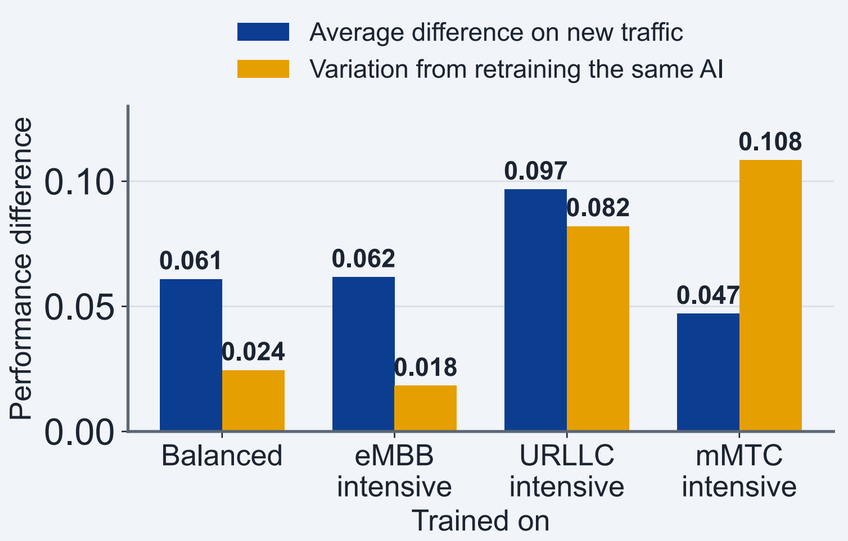
Difference in splits
Measured with a statistical formula of how different the splits are

5 Results

Performance loss when an AI is moved to a different environment



Performance difference from new environment, compared with retraining the same AI



6 Conclusions

- An AI trained on **balanced environments** performs as well as one retrained for each environment, so **there is no need to train a new model** every time the environment changes.
- Balanced agent transfers well across different environment conditions.

Contributions

Mark Zhao
Built the simulator and RL agent; chose the parameters and assumptions

Stanley Findsen
Led the literature review and linked our work to current industry practice